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Project Report: Modeling Demand Curve for Non-Traditional Patterns

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Abstract

In economic theory, the relationship between price and quantity demanded is classically captured by the law of demand, which posits an inverse relationship: higher prices generally lead to lower demand, resulting in a negatively sloped demand curve. However, this traditional model does not fully account for complex consumer behaviors and market dynamics that shape demand in modern economies. This project investigates alternative approaches to modeling demand, particularly for products where price changes do not yield typical demand responses. We explore a generalized non-linear regression model that includes weighted variables representing price and non-price factors, such as brand strength and distribution influence, to provide a more accurate depiction of market behavior.

Our approach employs constrained regression to ensure adherence to economically meaningful principles while permitting flexibility in demand patterns. Specifically, we apply constraints to enforce a negative slope on primary demand drivers where appropriate, thus aligning with traditional economic theory, while allowing deviations in cases of unique consumer preferences or atypical elasticity. By incorporating interactions between price and brand characteristics and considering potential non-linear relationships for primary price variables, our method improves predictive accuracy and provides a robust tool for analyzing markets with intricate pricing dynamics. The findings of this study demonstrate significant enhancements in demand prediction accuracy across various product categories, underscoring the practical value of our approach in economic analysis and demand forecasting for markets exhibiting complex consumer behavior.

1 Introduction

The relationship between price and quantity has long been a foundational concept in economics, underpinning the law of demand, which asserts that higher prices reduce consumer demand, yielding an inverse relationship between the two. This principle, essential to both microeconomic and macroeconomic analysis, is critical for setting effective pricing strategies, managing production levels, and optimizing supply chain logistics. The accompanying diagram (see Figure 1) illustrates this traditional downward-sloping demand curve, where price decreases as quantity increases, visually reinforcing the inverse relationship between the two.

However, modern markets exhibit substantial variability in consumer behavior, brand loyalty, and psychological factors that often violate traditional demand assumptions. Products influenced by perceived quality, exclusivity, and branding may not follow the classical downward-sloping demand curve, presenting challenges for companies aiming to accurately forecast demand.

In recent years, economic models have increasingly incorporated behavioral and psychological insights, revealing that consumer behavior is far from uniform. Some products, for instance, may experience stable or even increased demand at higher price points, as seen in luxury goods where higher prices may reinforce perceived exclusivity or quality. This deviation from the classical model, driven by factors such as brand

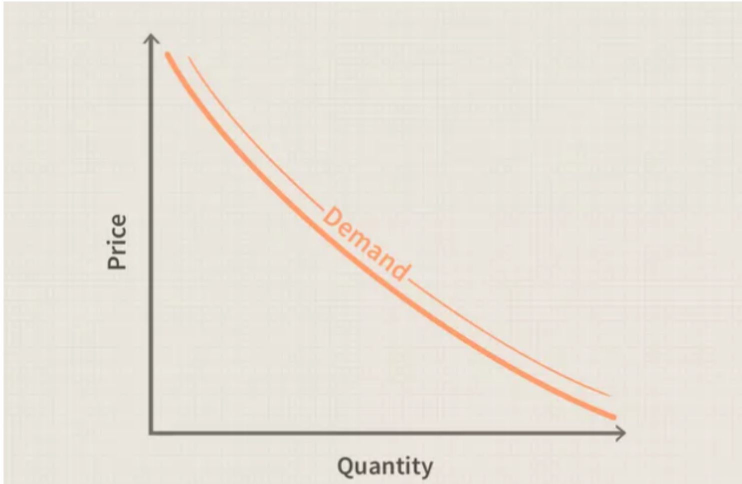

Figure 1: Classical demand curve showing the inverse relationship between price and quantity.

perception, peer influence, and consumer loyalty, underscores the need for a flexible modeling approach that captures both price sensitivity and non-linear demand behaviors.

This study introduces a non-linear regression model enhanced with weighted variables for price factors and product attributes, providing a comprehensive framework to analyze demand in markets with complex pricing and consumer dynamics. By implementing constraints in our regression model, we align the demand function with economic theory, ensuring that the primary price variable adheres to a negative slope requirement while allowing additional product features to influence demand in ways that reflect real-world deviations. Our model integrates price elasticity, marginal utility, and behavioral indicators, offering a more granular analysis of demand fluctuations.

The significance of this approach lies in its adaptability to a wide range of consumer behaviors across market segments, addressing the limitations of linear models. Traditional regression approaches may overlook nuanced price-quantity relationships that emerge in competitive and differentiated markets. By allowing for polynomial functions in key price-related features, our methodology provides a non-linear alternative that can accommodate shifts in demand elasticity and consumer behavior without compromising theoretical soundness.

This work is particularly relevant in contemporary economics, where rapid changes in consumer preferences are driven by social, economic, and psychological influences. With growing interest in adaptive models that can account for fluctuating market dynamics, this project contributes a flexible, data-driven tool for forecasting demand. Our findings indicate that this approach not only enhances the accuracy of demand predictions but also enriches the strategic insights available to economists and businesses alike, supporting more informed decision-making in complex, evolving markets.

2 Literature Survey

This section explores relevant literature on the relationship between price and quantity, consumer surplus, price elasticities, and regression models, which are foundational to our analysis of price-demand dynamics and consumer behavior.

2.1 Consumer and Producer Surplus

The paper by Choi [Choi2021] examines the theory of consumer and producer surplus, addressing how price variations affect consumer satisfaction and producer profit margins. Choi's analysis suggests that positive demand typically leads to an increase in consumer surplus, while a negative supply can reduce producer surplus. This insight emphasizes the impact of price fluctuations on consumer and producer welfare, which is directly relevant to our analysis, as our model aims to capture the sensitivity of demand to price changes.

2.2 Price Elasticity and Demand Response

The study by Hirth, Khanna, and Ruhnau [HirthKhannaRuhnau2022] focuses on the short-term price elasticity of German electricity demand, revealing how consumers adjust their demand in response to immediate price changes. This work is particularly relevant for understanding price elasticity, defined as the percentage change in quantity demanded resulting from a one-percent change in price. The findings provide a basis for incorporating elasticity considerations in our model, helping us estimate the degree to which changes in price affect consumer behavior in the short term.

2.3 Historical Perspectives on Consumer Surplus

Dooley's research [Dooley1983] provides a historical examination of consumer surplus, critiquing Marshall's approach to this concept and exploring alternative models that assess consumer satisfaction based on price changes. Dooley's analysis highlights critical debates about how to accurately measure consumer surplus and its relevance to economic welfare. These debates have influenced our approach to model design, particularly in the consideration of non-linear relationships between price and quantity, as consumer surplus is inherently affected by the elasticity of demand.

2.4 Empirical Studies on Demand Elasticities

The comprehensive study by Rosenthal-Kay, Traina, and Tran [RosenthalKayTrainaTran2021] analyzes "Seven million demand elasticities" across various sectors, providing a robust empirical basis for understanding how price responsiveness varies by industry. The study's findings underscore that demand elasticity can vary significantly depending on the product type, market structure, and time frame. This insight guided our decision to introduce polynomial terms for certain price-related features, allowing us to more closely mimic real-world, non-linear consumer responses to price changes.

2.5 Application of Regression Models in Consumer Behavior Analysis

The research reviewed also included various regression models, which have been widely used to analyze real-life consumer behavior by modeling non-linear relationships and allowing for flexibility in interpreting complex interactions between price, competition, and demand. This understanding led to our choice of polynomial regression in Approach 2, which offers a more realistic depiction of the price-quantity relationship, capturing the subtle and non-linear effects of price on demand.

2.6 Summary of Insights from Literature

Overall, the literature provides a multi-faceted understanding of price-demand relationships, elasticity, and consumer surplus. These studies informed our model choices and motivated the inclusion of polynomial terms and constraint optimization in our model. By applying these insights, our approach reflects the complexity of consumer demand, including the effects of price and competition, while adhering to economic principles observed in prior research.

References

[1] Choi, H. (2021). The Profitable Theory of Consumers' and Producers' Surplus: Positive Demand, Negative Supply?. Advisor, Chung-Hua Institution for Economic Research.

[2] Hirth, L., Khanna, T., Ruhnau, O. (2022). The (very) short-term price elasticity of German electricity demand. ZBW - Leibniz Information Centre for Economics, Kiel, Hamburg. Available at: `http://hdl.handle.net/10419/249570`

[3] Dooley, P. C. (1983). Consumer's Surplus: Marshall and His Critics. *The Canadian Journal of Economics / Revue canadienne d'Economique*, 16(1), 26-38. Published by Wiley on behalf of the Canadian Economics Association. Available at: `https://www.jstor.org/stable/134973`

[4] Rosenthal-Kay, J., Traina, J., Tran, U. (2021). Seven million demand elasticities. Leibniz Information Centre for Economics.

3 Data and Data Preparation

In this project, we utilized real-world data from multiple sectors. To maintain confidentiality, details of specific firms are omitted. The dataset primarily includes:

- **Numerical Columns:** The key numerical variables are *Price* and *Quantity*.
- **Categorical Columns:** Various categorical fields such as *PPG* (Promoted Product Group), *PackType*, *Brand*, *Channel*, *Market*, and sometimes *Distribution*.

Since categorical data cannot be directly applied to models, we performed feature engineering to create new variables that could better capture the underlying relationships. For our analysis, we primarily focused on *PPG/PackType* groups and conducted a detailed examination across brands within these groupings.

3.1 Feature Engineering

To capture the behaviors of markets and consumers in different areas, we created the following features:

1. **Price (PPL)**
   - **Description:** Price per Liter (PPL) serves as a critical factor in analyzing the performance of any brand, pack, or variant within a category.

- **Purpose:** Following the demand curve principle, an increase in price typically results in decreased volume, reflecting price sensitivity.
- **Method:** We apply a cubic function to *Price* to capture non-linear relationships between price and volume, which helps uncover complex consumer responses and deeper insights into price sensitivity.

2. **Competitive Pricing**
   - **Cannibalization Price:** Measures how one product's price affects the volume of another within the same brand or pack, capturing self-competition effects within a brand's offerings.
   - **Direct Competition Price:** Represents the effect of other competing brands offering similar products in the same category or *PackType* on volume, which helps understand the direct market competition.
   - **Indirect Competition Price:** Reflects how products in different categories, which serve similar consumer needs, influence volume—helping to capture cross-category effects.
3. **Seasonality, Trend, and Category Volume**
   - **Seasonality:** Calculated as the summed volume across all channel for a specific month or week , allowing insights into demand patterns over different time spans (monthly, quarterly, or yearly, depending on data).
   - **Trend:** The trend is derived by counting the data points in the time series, revealing long-term directional movements in demand.
   - **Category Volume:** This metric represents the total volume for a specific month or week across channels or markets. It helps assess the overall demand across various distribution channels.
4. **Distribution**
   - **Cannibalization Distribution:** Represents the impact of one product reducing sales of another within the same brand or pack, showing internal brand competition.
   - **Direct Competition Distribution:** Reflects the competitive pressure from similar products offered by other brands, helping assess competitive brand availability.
   - **Indirect Competition Distribution:** Represents the effect of products serving similar needs in different categories or packs, offering a broader view of market competition.
5. **Up-Down Index and Category Price Index**
   - **Up-Down Index:**
     - **Objective:** To quantify consumer behavior shifts over time with respect to price, assessing if consumers are moving towards costlier or cheaper products.
     - **Calculation:**
       (a) First, pivot the data on *Price* and *Volume* for each brand/variant in specific PPG/PACKTYPE groups.
       (b) Calculate average prices weighted by volume, then sum across brands to derive a single measure showing overall price behavior across brands.
     - **Interpretation:** A continuous curve that shows whether consumers are gravitating towards higher- or lower-priced products over time, helping to track market shifts.
   - **Category Price Index:**
     - **Goal:** To understand if the average price within a category is higher or lower relative to a baseline over time.

  - **Method:** Weighted by volume across PPGs, the price index normalizes the price behavior in a category, giving a comparative view over time.
  - **Usage:** Dividing the Up-Down Index by the Category Price Index reveals price movements, indicating if the category's price level is trending upwards or downwards.

6. **Brand Value and Interaction Terms**
   - **Brand Value:**
     - **Description:** Brand-specific factors encoded as binary values (0 or 1) that highlight brand distinctions.
   - **Interaction Terms:**
     - **Purpose:** By creating interaction terms between *Price* and one-hot encoded brand features, we capture brand-specific price sensitivity. This enables the model to reveal the unique impact of price changes on each brand.

Each feature contributes to a comprehensive view of consumer and market behavior, allowing us to model how price, competition, seasonality, and brand dynamics interact to drive sales volume. These engineered features are essential for capturing complex market trends, helping us understand and predict consumer responses more accurately.

4 Approach and Methodology

In general, we can represent the regression model as follows:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n \tag{1}$$

where y is the dependent variable (sales volume) and x_i represents the various independent variables or features, with their respective coefficients β_i.

4.1 Approach 1: Linear Model for PPG-Level Analysis by Brand

In our case, the model includes price sensitivity, competition, seasonality, trend, and brand interaction terms. Given that the model operates on the PPG level, we consider three PPG categories (Small, Medium, and Large) and four brands (A, B, C, D). The equation for the sales volume Q can be written as:

$$\begin{aligned}
Q = \beta_0 &+ \beta_1 \, \text{Price (PPL)} + \beta_2 \, \text{Price (Cannibalization)} + \beta_3 \, \text{Price (Direct Competition)} + \beta_4 \, \text{Price (Indirect Competition)} \\
&+ \beta_5 \, \text{Distribution (Distance)} + \beta_6 \, \text{Distribution (Cannibalization)} + \beta_7 \, \text{Distribution (Direct Competition)} + \beta_8 \, \text{Distribution (} \\
&+ \beta_9 \, \text{Seasonality} + \beta_{10} \, \text{Trend} + \beta_{11} \, \text{Up-Down Index} + \beta_{12} \, \text{Category Price Index} \\
&+ \beta_{13} \, \text{Category Volume} + \beta_{14} \, \text{Brand A (One-Hot Encoded)} + \beta_{15} \, \text{Brand B (One-Hot Encoded)} \\
&+ \beta_{16} \, \text{Brand C (One-Hot Encoded)} + \beta_{17} \, \text{Brand D (One-Hot Encoded)} \\
&+ \beta_{18} \, \text{Price (PPL)} \times \text{Brand A} + \beta_{19} \, \text{Price (PPL)} \times \text{Brand B} \\
&+ \beta_{20} \, \text{Price (PPL)} \times \text{Brand C} + \beta_{21} \, \text{Price (PPL)} \times \text{Brand D} + \epsilon
\end{aligned}$$

where:

- Q: The dependent variable, representing the predicted sales volume.
- $\beta_0, \beta_1, \ldots, \beta_{21}$: Coefficients representing the strength of each feature's impact on sales volume.
- ϵ: The error term, capturing any variation not explained by the model.

Each coefficient β_i reflects the strength of each feature's impact on the sales volume Q. The error term ϵ accounts for any variations not explained by the model.

*

Constraint Optimization for Regression Model

To solve this linear regression problem with specific constraints on the coefficients, we applied a **constrained optimization** approach. Here's a summary of the constraints we used for each type of feature, particularly focusing on the pricing variables to capture the required relationships:

- **Price (PPL)**: The coefficient for *Price (PPL)*, represented by β_1, was constrained to be **negative**. This reflects the intuitive expectation that, as the price per liter increases, the sales volume Q would generally decrease.
- **Price (Cannibalization), Price (Direct Competition), and Price (Indirect Competition)**: The coefficients for these terms—β_2, β_3, and β_4—were constrained to be **positive**. This captures the assumption that competitive or cross-category effects should have a positive impact on the sales of the main brand, as increases in competitors' prices make our product relatively more attractive.

To implement these constraints, we used a **constrained optimization algorithm** (such as Sequential Least Squares Quadratic Programming, SLSQP), which allows us to set boundaries on specific coefficients during the model training. This approach ensured that the model adhered to these business-driven assumptions, providing interpretable and meaningful coefficients that align with economic theory and practical insights.

So, the optimization problem can be formulated as follows:

$$\text{Minimize: } \sum (Q - \hat{Q})^2$$

Subject to:

$$\begin{aligned} &\beta_1 \leq 0 && \text{(for Price (PPL) to be negative)} \\ &\beta_2, \beta_3, \beta_4 \geq 0 && \text{(for other price terms to be positive)} \end{aligned}$$

This constraint-based approach allowed us to effectively capture the expected market behavior while fitting the model to the data.

4.2 Approach 2: Polynomial Regression with Degree 3 for Price (PPL)

In the second approach, we introduce **non-linearity for the Price (PPL) feature** by using a polynomial of degree 3. This adjustment acknowledges that the relationship between the price per liter (PPL) and sales volume may not be strictly linear; instead, it may exhibit diminishing or accelerating effects at different price levels.

By applying a polynomial transformation only to the *Price (PPL)* feature, we allow for greater flexibility in modeling the influence of price changes while maintaining a linear relationship for other price-related and non-price features. The remaining features, including "Price (Cannibalization)," "Price (Direct Competition)," and "Price (Indirect Competition)," retain their linear effects to reflect simpler, more direct influences on sales.

The modified regression model with polynomial terms for *Price (PPL)* is expressed as:

$$
\begin{aligned}
Q = {} & \beta_0 + \beta_1 \,\text{Price (PPL)} + \beta_2 \,\text{Price (PPL)}^2 + \beta_3 \,\text{Price (PPL)}^3 \\
& + \beta_4 \,\text{Price (Cannibalization)} + \beta_5 \,\text{Price (Direct Competition)} + \beta_6 \,\text{Price (Indirect Competition)} \\
& + \beta_7 \,\text{Distribution (Distance)} + \beta_8 \,\text{Distribution (Cannibalization)} + \beta_9 \,\text{Distribution (Direct Competition)} \\
& + \beta_{10} \,\text{Distribution (Indirect Competition)} + \beta_{11} \,\text{Seasonality} + \beta_{12} \,\text{Trend} \\
& + \beta_{13} \,\text{Up-Down Index} + \beta_{14} \,\text{Category Price Index} + \beta_{15} \,\text{Category Volume} \\
& + \beta_{16} \,\text{Brand A (One-Hot Encoded)} + \beta_{17} \,\text{Brand B (One-Hot Encoded)} \\
& + \beta_{18} \,\text{Brand C (One-Hot Encoded)} + \beta_{19} \,\text{Brand D (One-Hot Encoded)} \\
& + \beta_{20} \,\text{Price (PPL)} \times \text{Brand A} + \beta_{21} \,\text{Price (PPL)} \times \text{Brand B} \\
& + \beta_{22} \,\text{Price (PPL)} \times \text{Brand C} + \beta_{23} \,\text{Price (PPL)} \times \text{Brand D} + \epsilon
\end{aligned}
$$

4.2.1 Constraints for Polynomial Model

The constraints on the coefficients remain aligned with the business requirements set in the first approach, with additional considerations for the non-linear Price (PPL) terms:

- **Price (PPL)**: The coefficients β_1, β_2, and β_3 for *Price (PPL)* are constrained to maintain an overall **negative influence on sales volume** as Price (PPL) increases. This captures the assumption that higher product prices generally reduce demand.

- **Price (Cannibalization), Price (Direct Competition), and Price (Indirect Competition)**: These coefficients remain **positive**, indicating that higher prices of competing products could positively influence the sales of our product due to reduced competition.

By incorporating non-linearity only in the primary price feature, *Price (PPL)*, we achieve a model that balances flexibility in capturing complex price dynamics with interpretability and control over the other price-related factors.

4.3 Approach 3: Custom Ridge Regression with PPL Transformation

In this approach, we implemented a Custom Ridge Regression model designed to handle brand-level sales volume predictions while accounting for price sensitivity, seasonality, and brand-specific interactions. Key transformations applied to the primary price variable (`PPL`) aim to better capture non-linear relationships across different brands and channels.

Model Improvements Over Previous Approaches

The following improvements distinguish this approach from the previous two:

- **Transformation of Price Variable (PPL)**: We apply different transformations to the `PPL` variable to enhance the model's flexibility and capture non-linear relationships. Specifically, we tested reciprocal, logarithmic, exponential, and linear transformations, selecting the one with the lowest MAPE.

- **Interaction Terms for Brand and PPL**: By introducing interaction terms between transformed `PPL` values and encoded brand categories, the model adapts to varying price sensitivities across brands.

- **Constraints and Regularization**: We utilize L2 regularization (Ridge) along with constraints on feature weights, improving model stability and preventing overfitting.

Methodology

The workflow for this approach is detailed below:

Algorithm 1 Custom Ridge Regression with Brand Interaction Terms

```
Input:
    Dataset: df_ch_ppg
    Target variable: Volume
Initialize:
    Transformation functions: {reciprocal, log, exp_neg, linear}
    Predefined L2 penalty values
Procedure:
for transform_function ∈ {reciprocal, log, exp_neg, linear} do
    Transform the PPL feature using the transform_function
    One-hot encode the Brand column and create interaction terms with PPL_transformed
    Standardize all features
Optimization:
for l2_penalty in predefined range do
    Define the Ridge-regularized cost function
    Optimize parameters using the SLSQP algorithm with weight constraints
    Calculate training and test MAPE (Mean Absolute Percentage Error)
Update Best Parameters:
if training MAPE improves then
    Update the best parameters
    Record the minimum MAPE
Output:
    Best parameters
    Overall MAPE
    Brand-wise MAPE
```

5 Experiments

Evaluation

We evaluated the constrained model using MAPE metrics for both training and test datasets. This approach allowed us to maintain the interpretability of the demand curve while achieving acceptable prediction accuracy. Further experiments explore the effect of change of other functions and regularization on model performance.

5.1 Linear Ridge Regression

Our initial approach involved fitting the model using simple linear regression to predict `Volume` based on `PPG` and other pricing features. During this phase, we observed the following:

- For some `PPG/Pack` combinations, the model performed well, producing reasonable predictions.

- However, we encountered a critical issue: the calculated weights (β coefficients) for `PPG` were not always negative, which is essential for constructing a valid demand curve.

Imposing Constraints on Coefficients

To address this issue, we implemented a constrained regression approach as follows:

- A constraint was applied to ensure that the coefficient of `PPG` (`PPL`) is negative, reflecting the inverse relationship between price and demand.

- All other price-related coefficients were constrained to be positive to align with practical expectations.

- The optimization process aimed to minimize Mean Absolute Percentage Error (MAPE) while adhering to these constraints.

Linear Regression: Experimental Results

The results in Figure 3 summarize the MAPE values for training and testing across different channels, PPG, and brands. Additionally, Figure 2 shows the fit plot for testing data , allowing for a visual assessment of the model's performance on both datasets.

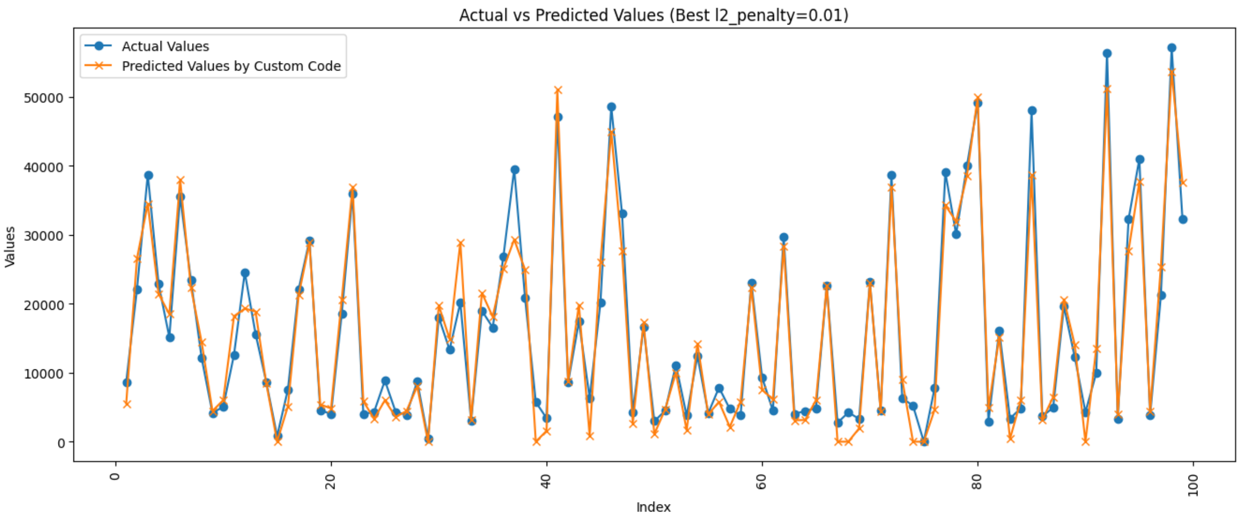

Figure 2: Fit Plot (Test Data) for Ridge Regression

Channel	PPG	Product Brands	Train_MAPE	Test_MAPE
Channel 1	PPG 1	['Brand A', 'Brand B', 'Other Brands']	14.82	17.84
Channel 1	PPG 2	['Brand A', 'Brand B', 'Private Label Std']	12.63	14.76
Channel 1	PPG 3	['Brand A', 'Brand C', 'Brand B', 'Brand D', 'Other Brands']	14.17	13.7
Channel 1	PPG 4	['Brand C', 'Brand B', 'Brand D', 'Other Brands']	11.84	12.55
Channel 1	PPG 5	['Brand B']	8.83	10.98
Channel 1	PPG 6	['Brand B']	8.43	10.3
Channel 2	PPG 5	['Brand A', 'Brand B']	16.0	11.06
Channel 2	PPG 4	['Brand A', 'Brand C', 'Brand B', 'Brand D']	15.92	17.06
Channel 2	PPG 3	['Brand A', 'Brand C', 'Brand B', 'Other Brands']	11.92	10.77
Channel 2	PPG 2	['Brand A', 'Brand B']	18.03	21.17
Channel 2	PPG 1	['Brand B']	7.38	8.78
Channel 3	PPG 5	['Brand A', 'Brand B']	10.2	11.66
Channel 3	PPG 4	['Brand A', 'Brand C', 'Brand B', 'Brand D', 'Brand E', 'Other Brands']	13.3	12.24
Channel 3	PPG 3	['Brand A', 'Brand C', 'Brand B', 'Brand D', 'Brand E', 'Other Brands']	11.99	11.04
Channel 3	PPG 2	['Brand A', 'Brand C', 'Brand B', 'Brand D', 'Brand E', 'Other Brands']	15.48	16.04
Channel 3	PPG 6	['Brand C', 'Brand B']	11.28	11.47
Channel 3	PPG 1	['Brand B', 'Other Brands']	14.64	12.12
Channel 4	PPG 4	['Brand A', 'Brand C', 'Brand B', 'Brand D', 'Other Brands']	12.71	13.87
Channel 4	PPG 2	['Brand A', 'Brand B', 'Brand D']	10.47	10.44
Channel 4	PPG 3	['Brand A', 'Brand C', 'Brand B', 'Brand D', 'Brand E', 'Other Brands']	16.57	15.59
Channel 4	PPG 1	['Brand B']	12.42	10.22
Channel 4	PPG 5	['Brand B']	12.57	17.09
Channel 4	PPG 6	['Brand B']	16.16	14.22

Figure 3: Results Table for Different Channels, PPGs, and Brands

5.2 Custom Ridge Regression with transformation

Reciprocal Transformation: Experimental Results

For the reciprocal transformation, no negative constraints were necessary on the PPL feature, unlike the linear transformation. This improved model performance, as reflected in the significantly lower MAPE for both the training and test sets, compared to the linear case. However, it should be noted that for certain combinations of PPG and brand levels, the reciprocal transformation did not perform as well, especially where the relationship between price and volume is more complex.

The corresponding fit plot and MAPE results for the reciprocal transformation are shown below:

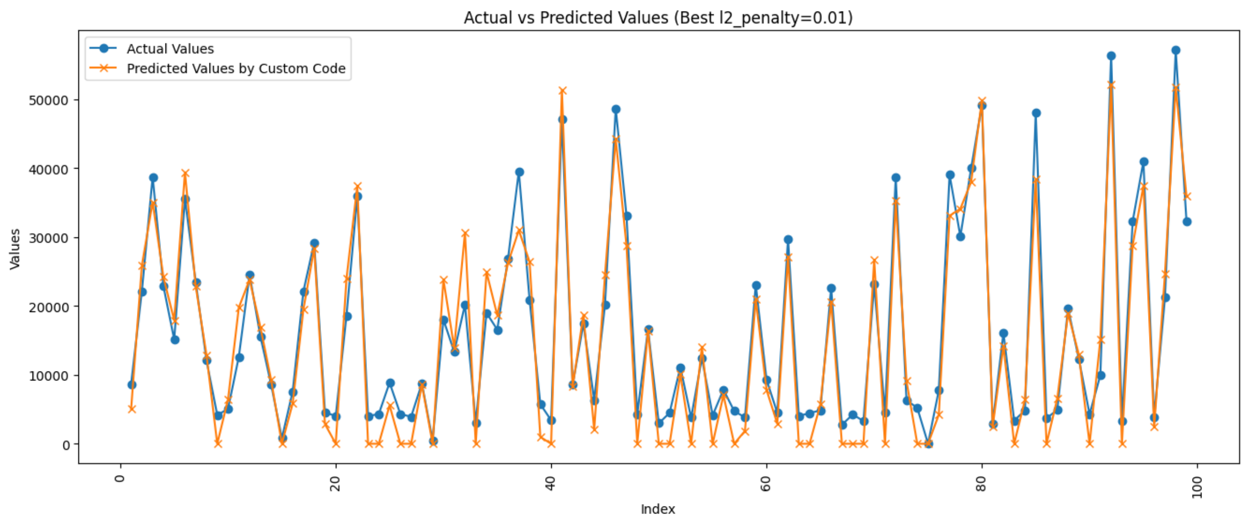

Figure 4: Fit Plot (Test Data) for Reciprocal Ridge Regression

Channel	PPG	Product Brands	Train_MAPE	Test_MAPE
Channel 1	PPG 1	['Brand A', 'Brand B', 'Other Brands']	15.94	18.65
Channel 1	PPG 2	['Brand A', 'Brand B', 'Brand C']	16.02	19.27
Channel 1	PPG 3	['Brand A', 'Brand C', 'Brand B', 'Brand D']	14.98	15.71
Channel 1	PPG 4	['Brand B', 'Brand C', 'Brand D']	14.45	14.71
Channel 1	PPG 5	['Brand B']	8.85	10.89
Channel 1	PPG 6	['Brand B']	8.17	10.21
Channel 2	PPG 5	['Brand A', 'Brand B']	14.51	10.18
Channel 2	PPG 4	['Brand A', 'Brand C', 'Brand B', 'Brand D']	18.31	18.85
Channel 2	PPG 2	['Brand A', 'Brand B']	11.23	10.02
Channel 2	PPG 3	['Brand A', 'Brand C', 'Brand B', 'Other Brands']	19.5	22.67
Channel 2	PPG 6	['Brand B']	7.4	8.82
Channel 3	PPG 5	['Brand A', 'Brand B']	11.75	12.42
Channel 3	PPG 4	['Brand A', 'Brand C', 'Brand B', 'Brand D']	14.29	13.1
Channel 3	PPG 2	['Brand A', 'Brand C', 'Brand B', 'Brand D']	12.8	12.12
Channel 3	PPG 3	['Brand A', 'Brand C', 'Brand B', 'Brand D', 'Other Brands']	16.46	16.43
Channel 3	PPG 6	['Brand C', 'Brand B']	11.93	11.05
Channel 3	PPG 1	['Brand B', 'Other Brands']	14.92	11.96
Channel 4	PPG 4	['Brand A', 'Brand C', 'Brand B', 'Brand D']	12.88	13.38
Channel 4	PPG 2	['Brand A', 'Brand B', 'Brand C']	11.06	11.04
Channel 4	PPG 3	['Brand A', 'Brand C', 'Brand B', 'Brand D', 'Other Brands']	18.66	18.73
Channel 4	PPG 1	['Brand B']	12.14	9.91
Channel 4	PPG 5	['Brand B']	13.12	16.91
Channel 4	PPG 6	['Brand B']	15.63	13.29

Figure 5: Results Table for Reciprocal Transformation - MAPE across Channels, PPGs, and Brands

Negative Exponential Transformation: Experimental Results

In comparison to the reciprocal transformation, the negative exponential approach proved more effective for certain PPG and brand combinations. This transformation naturally captured the downward-sloping demand curve, which is particularly useful in situations where there is a clear inverse relationship between price and volume. The negative exponential transformation did not require any additional constraints on the coefficients, and it led to better predictive performance in cases where the reciprocal transformation struggled to provide accurate results.

For these cases, the negative exponential transformation resulted in a more robust model with lower MAPE values across several key channels and PPG levels, as shown in the fit plot and MAPE table below.

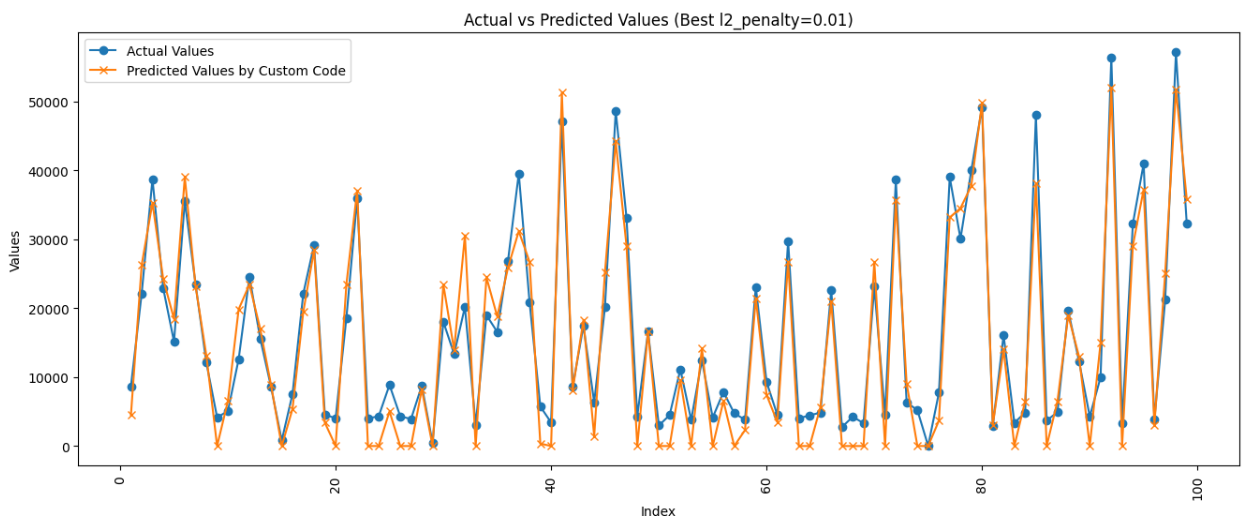

Figure 6: Fit Plot (Test Data) for Negative Exponential Ridge Regression

Channel	PPG	Product Brands	Train_MAPE	Test_MAPE
Channel 1	PPG 1	['Brand A', 'Brand B', 'Other Brands']	15.53	17.34
Channel 1	PPG 2	['Brand A', 'Brand B', 'Private Label Std']	15.92	19.44
Channel 1	PPG 3	['Brand A', 'Brand C', 'Brand B', 'Private Label Std']	15.2	15.84
Channel 1	PPG 4	['Brand C', 'Brand B', 'Private Label Std']	14.98	14.85
Channel 1	PPG 5	['Brand B']	9.94	11.57
Channel 1	PPG 6	['Brand B']	7.79	9.98
Channel 2	PPG 1	['Brand A', 'Brand B']	14.86	10.35
Channel 2	PPG 2	['Brand A', 'Brand C', 'Brand B']	18.54	19.98
Channel 2	PPG 3	['Brand A', 'Brand B']	11.67	10.32
Channel 2	PPG 4	['Brand A', 'Brand C', 'Brand B']	19.84	22.17
Channel 2	PPG 5	['Brand B']	7.47	8.88
Channel 3	PPG 1	['Brand A', 'Brand B']	11.9	12.53
Channel 3	PPG 2	['Brand A', 'Brand C', 'Brand B']	14.63	13.44
Channel 3	PPG 3	['Brand A', 'Brand B', 'Private Label Std']	12.98	12.1
Channel 3	PPG 4	['Brand A', 'Brand C', 'Brand B', 'Private Label Std']	17.79	17.95
Channel 3	PPG 5	['Brand C', 'Brand B']	11.78	11.0
Channel 3	PPG 6	['Brand B']	14.92	11.95
Channel 4	PPG 1	['Brand A', 'Brand B']	12.98	13.52
Channel 4	PPG 2	['Brand A', 'Brand B']	10.91	10.79
Channel 4	PPG 3	['Brand A', 'Brand C', 'Brand B']	19.59	19.08
Channel 4	PPG 4	['Brand A', 'Brand B']	12.14	9.98
Channel 4	PPG 5	['Brand A', 'Brand B']	13.32	16.95
Channel 4	PPG 6	['Brand B']	15.37	13.17

Figure 7: Results Table for Negative Exponential Transformation - MAPE across Channels, PPGs, and Brands